

Bots and how to catch them

Detekcija AI-generiranih Reddit komentara

ReinFERcement learning
STEM Games 2026 – Mathematics arena x Erste

Društvene mreže – danas više od 5,4 milijarde korisnika

Značaj dio prometa čine botovi (good i bad)

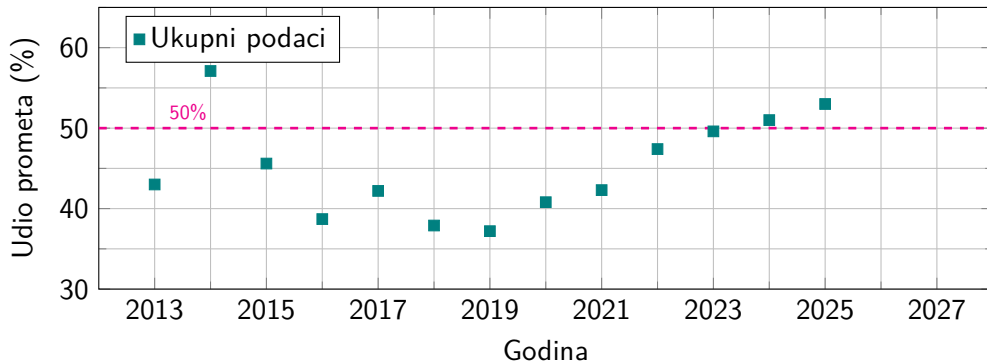
Problem bad botova – ozbiljna društvena prijetnja

Spam, phishing, manipulacija raspravama i umjetno povećanje angažmana

Dead internet theory

Ukupan udio botova na webu

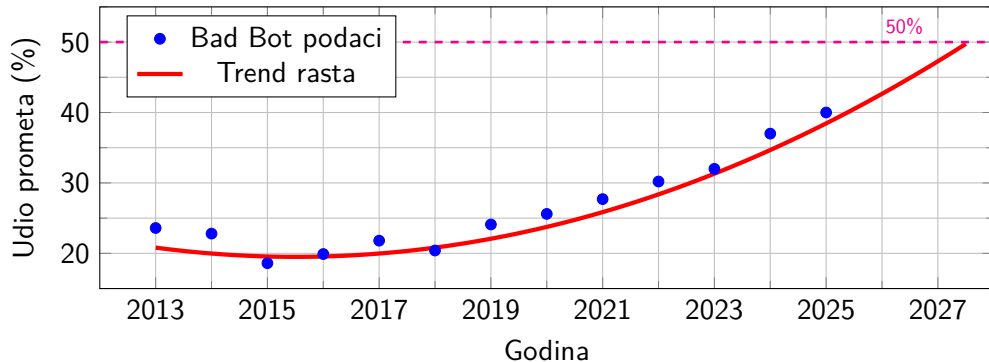
Ukupan udio botova (Bad + Good Bots)



Izvor: Statista, Human and bot web traffic share

Uspon “bad bot” prometa

Kvadratna aproksimacija rasta “Bad Bot” prometa



Izvor: Statista, Human and bot web traffic share

Ključne metrike bot prometa

Kategorija	Metrika	Vrijednost
Promet	Udio botova u ukupnom prometu	53%
	Udio ljudskog prometa	47%
Sigurnost	Udio “loših” botova	40%
	Godišnji porast AI bot napada	12.5x
Infrastruktura	Bot napadi na API-je	27%
	Bot napadi na poslovnu logiku	21%
Metode	Botovi koji koriste Chrome za prikrivanje	41%
Industrija	Napadi na financijske usluge	24%
	Preuzimanje računa u financijama	46%

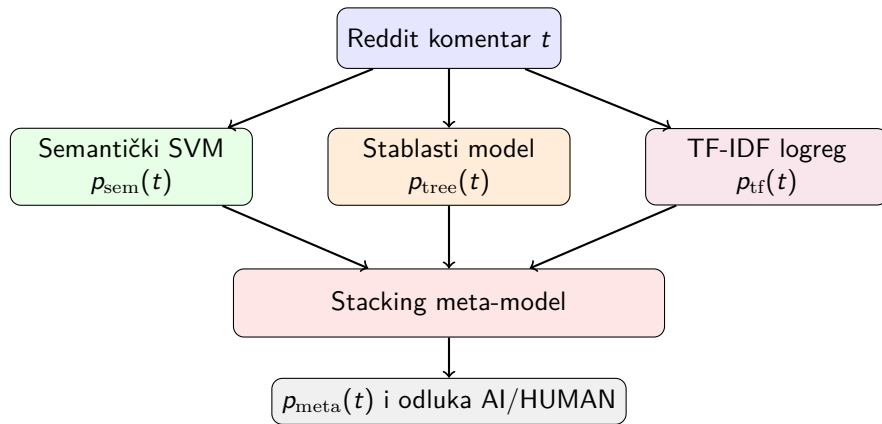
Izvor: Thales/Imperva, Bad Bot Report 2026

Detekcija i procjena vjerojatnosti AI generiranih komentara i objava

Stvaramo sustav koji spaja tri komplementarna pogleda u ansambl modela (ensemble learning)

Semantički pogled na značenje komentara, strukturni pogled na gradu rečenica i interpunkciju te leksički pogled na riječi, tokene i kratke znakovne obrasce

Završni stacking meta-model (kombinira 3 podsustava)



Tri komplementarna pogleda: značenje, struktura i lokalni tekstualni obrasci.

$$y = 1 \Leftrightarrow \text{AI}, \quad y = 0 \Leftrightarrow \text{HUMAN}$$

$$p(t) \approx P(y = 1 \mid t)$$

$$\hat{y} = \mathbf{1}\{p(t) \geq 0.5\}$$

Podsustav	Signal
SVM	semantički embedding
Stabla	20 strukturnih značajki
TF-IDF	1.56M rijetkih n -grama
Meta-model	kombinacija vjerojatnosti

Model 1: Semantički SVM

Prikaz komentara

$$\hat{x} = \frac{x}{\|x\|_2}, \quad z_j = \frac{\hat{x}_j - \mu_j}{\sigma_j}$$

Linearna odluka

$$f(z) = w^\top z + b$$

$$p_{\text{sem}}(t) \approx P(y = 1 \mid t)$$

Embedding model mapira tekst u prostor značenja.

LinearSVC razdvaja standardizirane embeddinge.

Vjerojatnost se dobiva Plattovom kalibracijom.

$$P(y = 1 \mid s) \approx \frac{1}{1 + \exp(As + B)}$$

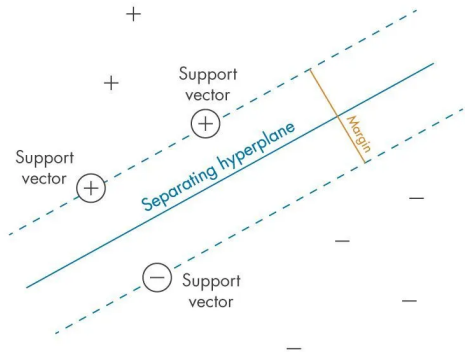
Semantički SVM: meka margina

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i$$

uz uvjete $y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0.$

$$\text{širina margine} = \frac{2}{\|\mathbf{w}\|_2}$$

Varijable ξ_i dopuštaju pogreške i primjere unutar margine;
parametar C određuje koliko ih model kažnjava.



Metrika	Dev
Točnost	0.90948
Preciznost	0.90341
Odziv	0.91283
F_1	0.90810
ROC AUC	0.96885

	$\hat{y} = 0$	$\hat{y} = 1$
$y = 0$	19549	2022
$y = 1$	1806	18912

Model 2: Stablasti model

Ulaz

$$x \in \mathbb{R}^{20}$$

Značajke: broj riječi i rečenica po odlomku, upitnici, apostrofi, zgrade, crtice, duljine rečenica, prijelazne riječi i brojčane reference.

Ansambl stabala

$$F_M(x) = \beta_0 + \eta \sum_{m=1}^M T_m(x)$$

$$p_{\text{tree}}(x) = \frac{1}{1 + e^{-F_M(x)}}$$

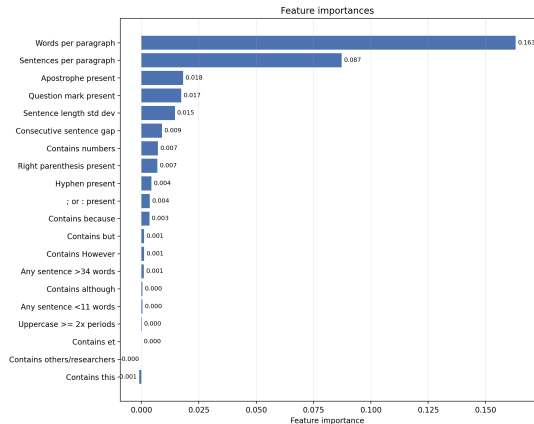
HistGradientBoostingClassifier: dubina 5, stopa učenja 0.06, 300 iteracija.

Stablasti model: heurističke značajke



Motivacija značajki: rečenična i odlomkovna struktura te diskurzivni signali. Izvor: PMC10328544, odjeljak S2.

Stablasti model: najvažnije značajke



Najveći doprinos: broj riječi po odlomku, broj rečenica po odlomku, apostrofi, upitnici i varijabilnost duljine rečenica.

Stablasti model: rezultati

Metrika	Dev
Točnost	0.90487
Uravnotežena točnost	0.9051
Preciznost	0.89214
Odziv	0.91664
F_1	0.90423
ROC AUC	0.96696

	$\hat{y} = 0$	$\hat{y} = 1$
$y = 0$	19275	2296
$y = 1$	1727	18991

Model 3: TF-IDF logistička regresija

TF-IDF

$$\text{tf}(t, g) = 1 + \log c_{t,g}$$

$$\text{idf}(g) = \log \frac{1 + N}{1 + \text{df}(g)} + 1$$

$$\tilde{x}_g(t) = \text{tf}(t, g) \text{idf}(g)$$

Dvije vrste n -grama

Riječi	1–2 grama
Znakovi	3–5 grama

$$x(t) = \begin{bmatrix} x_{\text{word}}(t) \\ x_{\text{char}}(t) \end{bmatrix} \in \mathbb{R}^{1557681}$$

TF-IDF logistička regresija: odluka i gubitak

$$s_c(x) = w_c^\top x + b_c$$

$$p_{\text{tf}}(t) = \frac{\exp(s_{\text{AI}}(x(t)))}{\exp(s_{\text{AI}}(x(t))) + \exp(s_{\text{HUMAN}}(x(t)))}$$

$$\min_w \sum_{i=1}^n \alpha_{y_i} [-y_i \log p_i - (1 - y_i) \log(1 - p_i)] + \lambda \|w\|_2^2$$

TF-IDF logistička regresija: rezultati

Metrika	Dev
Točnost	0.98085
Preciznost	0.98627
Odziv	0.97447
F_1	0.98033
ROC AUC	0.99780
Log-gubitak	0.06739
Brier	0.01626

	$\hat{y} = 0$	$\hat{y} = 1$
$y = 0$	21290	281
$y = 1$	529	20189

Stacking: kako spajamo tri modela

$$p_{\text{sem}}(t), \quad p_{\text{tree}}(t), \quad p_{\text{tf}}(t)$$

$$p_{\text{meta}}(t) \approx P(y = 1 \mid p_{\text{sem}}, p_{\text{tree}}, p_{\text{tf}})$$

Train ulazi za meta-model dobiveni su out-of-fold postupkom.

Finalni bazni modeli treniraju se na cijelom train skupu.

Meta-model je logistička regresija nad logitima i interakcijama.

$$\ell_i = \log \frac{p_i}{1 - p_i}, \quad p_i \in [10^{-5}, 1 - 10^{-5}]$$

$$\ell_1 = \text{logit}(p_{\text{sem}}), \quad \ell_2 = \text{logit}(p_{\text{tree}}), \quad \ell_3 = \text{logit}(p_{\text{tf}})$$

$$\phi(t) = [\ell_1, \ell_2, \ell_3, \ell_1 \ell_2, \ell_1 \ell_3, \ell_2 \ell_3, \ell_1^2, \ell_2^2, \ell_3^2]$$

$$p_{\text{meta}}(t) = \sigma \left(\beta_0 + \sum_{i=1}^3 \beta_i \ell_i + \sum_{i < j} \beta_{ij} \ell_i \ell_j + \sum_{i=1}^3 \gamma_i \ell_i^2 \right)$$
$$\sigma(u) = \frac{1}{1 + e^{-u}}, \quad \hat{y} = \mathbf{1}\{p_{\text{meta}}(t) \geq 0.5\}$$

Interakcije dopuštaju modelu da razlikuje slaganje i proturječje baznih podsustava.

Usporedba modela na razvojnom skupu

Model	Točnost	Prec.	Odziv	F_1	AUC	Log-gub.	Brier
Semantički SVM	0.90948	0.90341	0.91283	0.90810	0.96885	0.22646	0.06715
Stablasti model	0.90487	0.89214	0.91664	0.90423	0.96696	0.23115	0.07009
TF-IDF logreg	0.98085	0.98627	0.97447	0.98033	0.99780	0.06739	0.01626
Meta-model	0.98406	0.98788	0.97949	0.98366	0.99840	0.04564	0.01240

Meta-model je najbolji po svim navedenim metrikama.

Konfuzijske matrice na razvojnom skupu

Model	TN	FP	FN	TP
Semantički SVM	19549	2022	1806	18912
Stablasti model	19275	2296	1727	18991
TF-IDF logreg	21290	281	529	20189
Meta-model	21322	249	425	20293

Meta-model

	$\hat{y} = 0$	$\hat{y} = 1$
$y = 0$	21322	249
$y = 1$	425	20293

Meta-model smanjuje i lažno pozitivne i lažno negativne primjere u odnosu na najjači pojedinačni model.

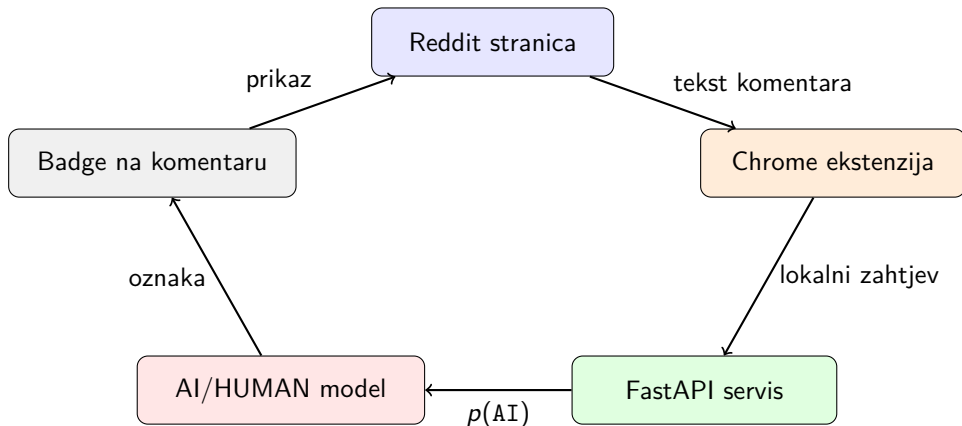
Test rezultat završnog modela

Točnost	Prec.	Odziv	F_1	AUC	Log-gub.	Brier
0.98406	0.98722	0.98016	0.98368	0.99842	0.04499	0.01226

$$p_{\text{meta}}(t) \in [0, 1]$$

Završni izlaz nije samo oznaka, nego procjena vjerojatnosti AI autorstva.

Arhitektura aplikacije



Ekstenzija samo prikazuje rezultat; klasifikacija se izvodi lokalno u FastAPI aplikaciji.

Demo ekstenzije: mogući AI komentar

Young men between the ages of 18 and 35 are...

reddit

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- Blog

another Right Wing position.

1

Ok-Childhood-8775 • 21d ago

The way you frame it like it is something bad. But I would argue it is better to care for something and at least try to change something instead of just going with the flow. But maybe you are right and everything truly is meaningless.

10

Aggravating-Math3794 • 21d ago

Possibly AI-like 84%

Wait, did it come off like that? I actually didn't mean to imply that having personal urges behind ideological beliefs is bad or that it makes the big scale ideologies meaningless - more like that it's natural for the personal drive to be there first, but a lot of people pretend like it's all about the big picture to hide their insecurities.

Like, I have plenty of political and moral beliefs as surely most people do, and I care about them a lot - my point was mainly about people who use ideologies and trends as a way to distract themselves from their deep issues and assert dominance rather than to understand self better, so it's really about one's self-awareness and intent rather than if having personal reasons to speak ideology is good or bad.

I hope I didn't explain it too messily.

5

GhostofBeowulf • 20d ago

I mean I feel personally attacked right now lol.

While I am not super boisterous irl, I do speak a lot and have strong feelings on politics. I also desire to invest myself into my local and state political process, so much so I am going back for a second bachelor's with a dual major in public administration and business/org management, and will probably some day run for office long after I finish grad school. But also I have a slightly covert intent, and that is developing, implementing and popularizing anarchist principles in our capitalist society to make a more fair, equal and just society. Basically take my education and use it to frame these anarchist/co-production/co-governance ideals as solutions to economic and market based externalities and inefficiencies, and show that these are the most efficient tools that yield the best, most equitable outcomes for all.

r/stupidpol • 2mo ago

Opinion: Most young people who identify as Left-wing these days have no idea what Lef...

338 upvotes · 223 comments

r/BlackPeopleTwitter • 4mo ago

Political parties are excuse for normal people to turn their bra...

22K upvotes · 390 comments

r/teenagers • 9d ago

Why do people always get mad when I tell them my political views?

2 upvotes · 18 comments

r/psychology

Psychology

A Reddit community for sharing and discussing science-based psychological material.

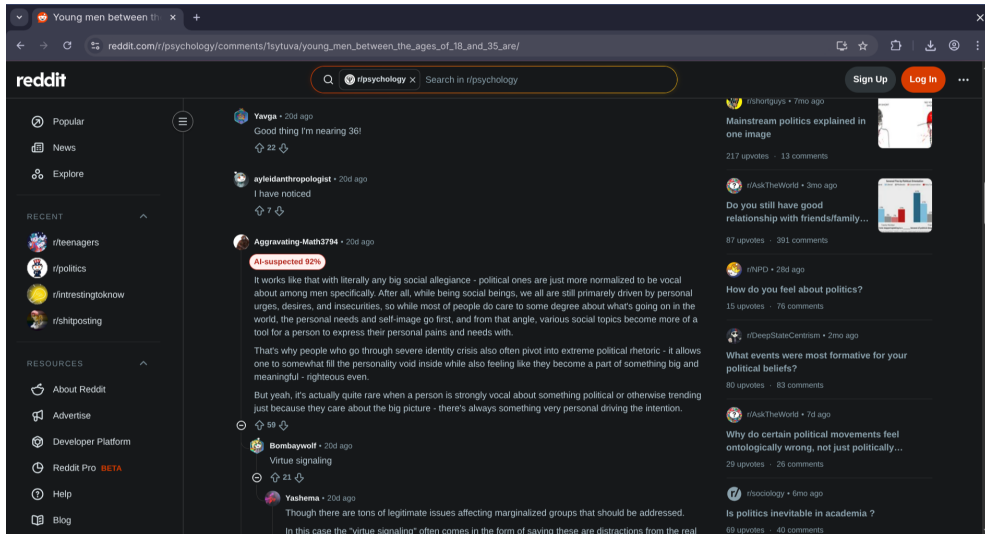
Public

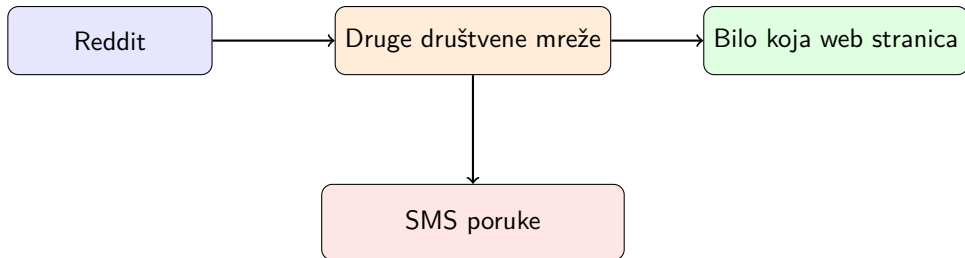
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Demo ekstenzije: visoka AI sigurnost





Isti princip: tekst $\rightarrow$ modeli $\rightarrow$ procjena rizika / AI autorstva.

Što bi trebalo za produkcijsko skaliranje

veći i raznolikiji dataset

jači modeli

jači hardver

testiranje na različitim platformama i jezicima

arhitektura ostaje ista

Koristimo točno tri bazna modela: semantički SVM, stablasti model i TF-IDF logističku regresiju.

Modeli gledaju različite signale: značenje, strukturu i lokalne tekstualne obrasce.

Stacking meta-model kombinira njihove vjerojatnosti preko logita, interakcija i kvadratnih članova.

Završni model postiže najbolji rezultat: dev accuracy 0.98406, test accuracy 0.98406.

Sustav procjenjuje vjerojatnost AI-generiranosti komentara, a ne dokazuje identitet autora.